

Kavish Agarwal

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Technology Leader | Enterprise Architect | Principal Engineer | Distributed Systems | Platform Modernization | Cloud & AI

EXECUTIVE PROFILE

Technology leader, enterprise architect, and engineering multiplier with 25+ years of experience building and modernizing mission-critical platforms across payments, financial services, healthcare, retail, utilities, and travel. Deep expertise in payments, real-time decisioning, fraud, and risk, with broader leadership across distributed systems, enterprise platforms, legacy modernization, cloud, AI, and engineering transformation. At Visa, progressed from individual ownership of critical modernization work to technical leadership of multi-year global platform programs, reusable enterprise decisioning foundations, and an interim Senior Director role. Known for multiplying organizational capability by developing technical leaders, mentoring distributed teams, creating reusable patterns and platforms, and enabling teams to deliver independently at scale.

CORE LEADERSHIP, ARCHITECTURE & TRANSFORMATION

Payments, Decisioning & Risk

Card & A2A payments; fraud/risk; authentication; real-time decisioning; business rules; ML integration

Leadership & Organizational Scale

Executive alignment; global team leadership; technical strategy; mentoring; engineering multiplier; inner sourcing

Enterprise & Platform Architecture

Distributed systems; platform strategy; microservices/SOA; event-driven architecture; reusable platforms; architecture governance

AI & Engineering Transformation

Agentic AI; LLM orchestration; workflow automation; human-in-the-loop systems; AI-enabled engineering; DevSecOps

Modernization & Cloud Transformation

Legacy modernization; cloud-native architecture; incremental decomposition; migration/coexistence; HA/DR; observability

Cross-Domain Experience

Payments; financial services; healthcare claims; retail/pharmacy; utilities; travel

PROFESSIONAL EXPERIENCE

Visa | Austin, TX

Chief Software Engineer | May 2015 – Present

- **Enterprise platform leadership:** Core technical leader shaping Visa's integrated/unified risk decisioning platform across issuing, acquiring, card, and A2A use cases; helped evolve reusable foundations from Risk Management as a Service into an enterprise platform supporting multiple fraud and risk products.
- **Real-time payments & A2A platform leadership:** Led the architecture and evolution of A2A real-time risk decisioning from initial service conception through production deployment, shaping reusable scoring, rules, orchestration, and regional deployment capabilities for global clients; later led the Single API architecture and Featurespace integration as part of the broader platform evolution.
- **Reusable decisioning foundations:** Primary designer/implementation lead for generalized orchestration and aggregation engines used across real-time, event, and offline flows, including enumeration-attack and velocity use cases; influenced rule management, rule publishing, and OPA-based decisioning architecture.
- **Platform scale & distribution:** Led architecture exploration, benchmarking, and POCs for distributing ~2M OPA business rules across multi-tenant clusters, evaluating bundle granularity, placement by tenant/volume, dedicated-vs-shared clusters, rebalancing, and directory/discovery-based traffic routing.
- **Global modernization:** Technical lead for multi-year modernization of Visa Risk Manager, a mission-critical risk platform serving ~1,300 banks and ~25,000 users; guided decomposition of a legacy Java/Struts/WebSphere monolith into independently deployable services while preserving shared business logic and existing data to reduce migration risk.
- **Large-scale platform delivery:** Led end-to-end system design and production readiness for 125+ servers across six application stacks, partnering with Production Release Engineering, infrastructure, middleware, cybersecurity, and DBA teams; designed active-active patterns, observability, failover, and session-continuity solutions.
- **Product & user migration:** Partnered with product management on an eight-month phased global migration strategy that moved ~25,000 users through pilot and expansion waves, using feedback-driven rollout and controlled coexistence with the legacy platform.

Visa | continued

- **Engineering & team leadership:** Guided 8+ distributed teams and 60+ engineers across North America, India, and Poland, working across product, data science, platform, security, operations, and architecture communities; mentored new teams, ran architecture/design reviews and office hours, and served in an interim Senior Director capacity with three managers reporting through the organization.
- **Architecture governance:** Helped establish inner-sourcing, deployment/support models, reusable platform principles, and AWG/ADR practices; drove design reviews with 50+ engineers and used structured ADRs to evaluate options, tradeoffs, non-functional requirements, and reference implementations.
- **Operational excellence & cloud economics:** Owned A2A production-excellence responsibilities for a platform targeting 99.999% availability; drove incident closure/postmortems, EKS/Go/Redis performance tuning, IaC and autoscaling, and cost optimization using right-sizing, Valkey-based ElastiCache, and Graviton-based compute.
- **Engineering transformation & AI adoption:** Helped move teams from waterfall/ClearCase toward Agile, Git, Jenkins CI/CD, automation, and performance testing; more recently created reusable AI skills/agents, codebase knowledge tooling, and AI-augmented micro-team patterns for engineering delivery.
- **Innovation:** Named inventor on U.S. Patent 11,880,845 for real-time account-level rule exclusion in real-time payment transaction processing.

PayPal

Member Technical Staff (MTS) | September 2011 – May 2015

- **Decision-platform modernization:** Recruited as a Blaze Advisor SME to modernize PayPal risk and fraud decisioning; built and validated a new implementation, then helped realign the program to a modernized IBM ODM platform when product feedback showed a better fit with existing business workflows and capabilities.
- **Risk scoring & technical leadership:** Individually led an early risk-scoring implementation on the decisioning platform in close partnership with data science, translating model logic into executable decisioning flows; continued enhancing the platform, evaluating rule/data technologies, standardizing engineering practices, and mentoring teams on decisioning architecture.

EARLIER EXPERIENCE

Cognizant Technology Solutions — Principal Consultant | 2008–2011

Cross-domain enterprise architect across insurance underwriting, mortgage servicing, healthcare, utilities, and data platforms; led target-state modernization design, guided distributed delivery and migration, and supported pre-sales through POCs and client presentations.

Wipro Technologies — Technical Consultant | 2004–2008

Progressed through Technical Lead, Project Manager, and Rules Architect roles; led FICO Blaze Advisor work for Medicare claims and retail-pharmacy modernization, trained teams, and shaped configuration-driven platforms for regulated domains.

GE Capital Services — Technical Consultant | 2003–2004

Built enterprise process-automation solutions for GE Power using PegaRULES Process Commander, applying rules and workflow platforms to business operations.

Virtusa — Senior Software Engineer | 2001–2003

Worked on retail supply-chain systems and PegaRULES Process Commander modernization, including expression-language parser/compiler components; an early foundation in reusable platform and productization thinking.

StreamSoft India — Earlier career experience

Built Java/Applet-based travel-reservation capabilities across airline, hotel, and car workflows, with IATA/Sabre exposure and early UK client delivery.

TECHNICAL SKILLS

Domains & decisioning: Payments, fraud/risk, healthcare claims, retail/pharmacy, process automation, real-time scoring, business rules, ML integration

Architecture & platforms: Distributed systems, event-driven architecture, microservices/SOA, real-time processing, active-active, HA/DR, platform modernization, governance

Cloud & infrastructure: AWS, EKS, Kubernetes, Kafka, Redis/Valkey, Helm, Docker, Lambda, EMR, API Gateway, DynamoDB, CI/CD, DevSecOps, observability

Engineering, rules & AI: Java, Go, gRPC, REST, Spring, Cassandra, PostgreSQL, Pega PRPC, FICO Blaze Advisor, IBM ODM, OPA, XGBoost, Agentic AI, LLM orchestration, multi-agent systems

EDUCATION, PROFESSIONAL EDUCATION & CERTIFICATIONS

Education: B.E., Electronics & Communications — Government Engineering College, Bhopal, India

Professional Education: MIT — Applied Generative AI for Digital Transformation • **MIT xPRO** — Designing and Building AI Products and Services

Stanford — Advanced Cyber Security Certificate • **UT Austin Executive Education** — Advocacy, Influence & Power

Certifications: TOGAF Certified • AWS Certified Cloud Practitioner • Pega PRPC Certified System Architect